

Your Name:

ID #:

Solution

Note: for all the problems, identify the limit as a number, $-\infty$, ∞ or DNE (does not exist)

1. Please keep in mind the list of functions in order of their rate of growth – quickest to slowest:

Factorial($x!$), **Exponential** ($4^x, e^x$), **Algebraic** ($x^4, x^{0.5}$), **Logarithm** ($\log_2 x, \ln x$)

Then find the following limits quickly without showing any work:

$$(a) \lim_{x \rightarrow \infty} \frac{x^3}{x!} = 0$$

$$(g) \lim_{n \rightarrow \infty} \frac{5^n + 2}{n^9 + 100} = \infty$$

$$(m) \lim_{x \rightarrow \infty} \frac{x^8 + \ln x}{3x^8 + 2} = \frac{1}{3}$$

$$(b) \lim_{x \rightarrow \infty} \frac{\sqrt[3]{x^4 + 2x^3 + 1}}{\ln x + 3} = \infty$$

$$(h) \lim_{n \rightarrow \infty} \frac{n^4 + 10 \ln n}{4\sqrt{n^8 + 3n + 100}} = \frac{1}{4}$$

$$(n) \lim_{x \rightarrow \infty} \frac{x^{10} + 2x^2 + 10}{32x^6 + 12} = \infty$$

$$(c) \lim_{x \rightarrow \infty} \frac{x^{10} + 4x^7 + 10}{32x^{12} + 10} = 0$$

$$(i) \lim_{n \rightarrow \infty} \frac{-n^4 + 3n^2}{19n^3 + 3n^2 - 10} = -\infty$$

$$(o) \lim_{n \rightarrow \infty} \frac{n^5 - 10n^7 + \ln n}{n^3 + 8n^7} = -5/4$$

$$(d) \lim_{x \rightarrow \infty} \frac{5x + 1}{x^2 - 3x + 4} = 0$$

$$(j) \lim_{x \rightarrow -\infty} \frac{x^{120} + 20}{-x^{100} + e^x} = -\infty$$

$$(p) \lim_{x \rightarrow \infty} \frac{e^x}{x^3} = \infty$$

if $x \rightarrow +\infty$, then 0

$$(e) \lim_{x \rightarrow -\infty} \frac{2^x}{x^2} = 0$$

$$(k) \lim_{n \rightarrow -\infty} \frac{5^n + n^2}{-e^n + 1} = \infty$$

$$(q) \lim_{x \rightarrow -\infty} (-3x^3 - 4x + 5) = \infty$$

if $n \rightarrow +\infty$, then $-\infty$

$$(f) \lim_{x \rightarrow \infty} \frac{2^x}{x^2} = \infty$$

$$(l) \lim_{x \rightarrow -\infty} \frac{2x^3 - 2x + 5}{13x^3 - 5x + 13} = \frac{2}{13}$$

$$(r) \lim_{n \rightarrow \infty} \frac{n^5 - 10n^7 + \ln n}{n^3 + 8n^7} = \infty$$

2. Practice some common limits without showing the work:

$$(a) \lim_{x \rightarrow -\infty} e^x = 0$$

$$(d) \lim_{x \rightarrow -\infty} e^{-x} = \infty$$

$$(g) \lim_{x \rightarrow -\infty} e^{\frac{1}{x}} = e^0 = 1$$

$$(b) \lim_{x \rightarrow 0} e^x = e^0 = 1$$

$$(e) \lim_{x \rightarrow 0^+} e^{\frac{1}{x}} = \infty$$

$$(h) \lim_{x \rightarrow \infty} e^{\frac{1}{x}} = e^0 = 1$$

$$(c) \lim_{x \rightarrow -\infty} x \ln x \text{ "No sense"}$$

$$(f) \lim_{x \rightarrow 0^-} e^{\frac{1}{x}} = 0$$

$$(i) \lim_{x \rightarrow \infty} \frac{1}{e^{-x}} = \infty$$

if $x \rightarrow +\infty$
then ∞

3. Practice more common limits without showing the work:

$$(a) \lim_{x \rightarrow \infty} \ln x = \infty \quad (e) \lim_{x \rightarrow \infty} \sin x = \text{DNE} \quad (i) \lim_{x \rightarrow \infty} \sin\left(\frac{1}{x}\right) = \sin 0 = 0$$

$$(b) \lim_{x \rightarrow 0^+} \ln x = -\infty \quad (f) \lim_{x \rightarrow \infty} (\sin x)e^{-x} = 0 \quad (j) \lim_{x \rightarrow \infty} x \ln \frac{1}{x} = -\infty$$

$$(c) \lim_{x \rightarrow -\infty} (e^{3x} - e^{2x}) = 0 \quad (g) \lim_{x \rightarrow \infty} x \sin x = \text{DNE} \quad (k) \lim_{x \rightarrow -\infty} x e^{\frac{1}{x}} = -\infty$$

if $x \rightarrow +\infty$, then ∞ if $x \rightarrow +\infty$, then $+\infty$

$$(d) \lim_{x \rightarrow \infty} \sin\left(\frac{x^2}{2x^2+x}\right) = \sin\left(\frac{1}{2}\right) \quad (h) \lim_{x \rightarrow \infty} \frac{1}{x} \sin x = 0 \quad (l) \lim_{x \rightarrow 0^-} x e^{\frac{1}{x}} = 0$$

if $x \rightarrow 0$, then 1

4. Show all you work for the following questions.

$$(a) \lim_{x \rightarrow 0} \frac{\sin(3x)}{4x} \quad \text{or} \quad \begin{aligned} &= \frac{3}{4} \lim_{x \rightarrow 0} \frac{\sin(3x)}{3x} \quad \text{L.H} \\ &= \frac{3}{4} \lim_{x \rightarrow 0} \frac{3 \cos(3x)}{4} \\ &= \frac{3}{4} \cdot 1 \\ &= \boxed{\frac{3}{4}} \end{aligned}$$

$$(b) \lim_{x \rightarrow \infty} x^{\frac{1}{x}} \quad \text{" } \infty^0 \text{ "}$$

$y = x^{\frac{1}{x}} \Rightarrow \ln y = \ln x^{\frac{1}{x}} \Rightarrow \ln y = \frac{1}{x} \ln x$

$$\lim_{x \rightarrow \infty} \ln y = \lim_{x \rightarrow \infty} \frac{\ln x}{x} \quad \text{L.H} \quad \lim_{x \rightarrow \infty} \frac{1}{x} = 0$$

$$\Rightarrow \lim_{x \rightarrow \infty} x^{\frac{1}{x}} = \lim_{x \rightarrow \infty} y = e^0 = \boxed{1}$$

$$(c) \lim_{x \rightarrow \infty} \left(\frac{1}{x}\right)^{\frac{1}{x}} \quad \text{" } 0^0 \text{ "}$$

let $y = \left(\frac{1}{x}\right)^{\frac{1}{x}}$, then $\ln y = \ln\left(\frac{1}{x}\right)^{\frac{1}{x}} = \frac{1}{x} \ln\left(\frac{1}{x}\right)$

$$\ln y = \frac{1}{x} \ln(x^{-1}) = -\frac{1}{x} \ln x = -\frac{\ln x}{x}$$

$$\lim_{x \rightarrow \infty} \ln y = \lim_{x \rightarrow \infty} \left(-\frac{\ln x}{x}\right) \quad \text{L.H} = 0$$

meanwhile, $\lim_{x \rightarrow \infty} \ln y = \ln(\lim_{x \rightarrow \infty} y) = 0$

$$\Rightarrow \lim_{x \rightarrow \infty} y = \lim_{x \rightarrow \infty} \left(\frac{1}{x}\right)^{\frac{1}{x}} = e^0 = \boxed{1}$$

$$(d) \lim_{x \rightarrow 1} \frac{5x^4 - 4x^2 - 1}{10x^3 - 9x^2} \quad \text{" } \frac{0}{0} \text{ "}$$

$$\text{L.H} \quad \lim_{x \rightarrow 1} \frac{20x^3 - 8x}{-1 - 27x^2}$$

$$= \frac{20 - 8}{-1 - 27} = \boxed{-\frac{3}{7}}$$

$$(e) \lim_{x \rightarrow -1} \frac{\sqrt{x+4} - 3}{x+1}$$

$$= \boxed{\text{DNE}} \quad \text{since}$$

$$\lim_{x \rightarrow (-1)^+} \frac{\sqrt{x+4} - 3}{x+1} = -\infty$$

$$\text{but } \lim_{x \rightarrow (-1)^-} \frac{\sqrt{x+4} - 3}{x+1} = +\infty$$

$$(f) \lim_{x \rightarrow \infty} \left(\frac{x}{x+1}\right)^x \quad \text{" } 1^\infty \text{ "}$$

$$y = \left(\frac{x}{x+1}\right)^x \Rightarrow \ln y = x \ln\left(\frac{x}{x+1}\right)$$

$$\lim_{x \rightarrow \infty} x \ln \frac{x}{x+1} = \lim_{x \rightarrow \infty} \ln\left(1 - \frac{1}{x+1}\right)$$

$$\text{L.H} \quad \lim_{x \rightarrow \infty} \frac{\frac{1}{1 - \frac{1}{x+1}} \cdot \left(-\frac{1}{(x+1)^2}\right)}{\left(-\frac{1}{x+1}\right)^2} = -1$$

$$\Rightarrow \lim_{x \rightarrow \infty} y = \boxed{e^{-1}}$$